

CHE 201 Spring 2020 HW 2

Roxanna Mendoza

TOTAL POINTS

85 / 100

QUESTION 1

1 Question 1 20 / 20

✓ - **0 pts** Correct

- **5 pts** Effect of friction is wrong or not answered.
- **5 pts** Calculation of constant force R is wrong.
- **5 pts** Calculation of the force of gravity is wrong.
- **5 pts** Energy balance is wrong.

QUESTION 2

2 Question 2 20 / 20

✓ - **0 pts** Correct

- **5 pts** Wrong calculation in b)
- **0 pts** Be more precise on calculation. Keep more significant digits.
- **5 pts** Wrong equation in b)
- **5 pts** Wrong calculation in a)
- **5 pts** Wrong equation in a)

QUESTION 3

3 Question 3 20 / 20

✓ - **0 pts** Correct

- **4 pts** Wrong sketch of the processes.
- **4 pts** Wrong answer in a)
- **4 pts** Wrong 1-2 work in b)
- **4 pts** Wrong 2-3 work in b)
- **0 pts** Please distinguish a positive or a negative sign to denote the energy transfer direction by work.
- **4 pts** Wrong 3-1 work in b)

QUESTION 4

4 Question 4 10 / 20

- **0 pts** Correct
- **5 pts** Wrong equation used in getting power
- **5 pts** Wrong calculation in getting power.
- ✓ - **5 pts** Wrong equation used in getting the speed

✓ - **5 pts** Wrong calculation in getting the speed

QUESTION 5

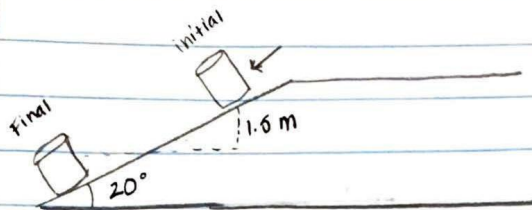
5 Question 5 15 / 20

- **0 pts** Correct
- **2.5 pts** Not enough current federal regulations.
- **2.5 pts** Not enough current regulations in state and local area.
- ✓ - **5 pts** Not enough comments on the effectiveness of services.
- **20 pts** Question 5 not answered.
- **5 pts** No current federal regulations.
- **2 pts** No reference added.
- **5 pts** No current regulations in state or local area.

Homework #2

Roxanna Mendoza

#1.



$$m = 0.4 \text{ kg}$$

$$R = 0.05 \text{ N}$$

$$g = 9.8 \text{ m/s}^2$$

$$W = \Delta KE + \Delta PE$$

$$W = F \times L$$

$$W = 0.05 \text{ N} \times \frac{1.5}{\sin 20^\circ}$$

$$W = 0.22 \text{ Joules}$$

Work is equal to the product of the force and the distance

$$PE = m \cdot g \cdot \Delta h = (0.4 \text{ kg}) (9.8 \text{ m/s}^2) (1.5 \text{ m}) = -5.88 \text{ Joules}$$

$$KE = W - PE \Rightarrow KE = 0.22 - (-5.88) = 6.1 \text{ Joules} \leftarrow \text{increasing}$$

- if the friction between the can and inclined surface were significant the kinetic energy will be decreased because there'll be some loss of energy due to the friction.

#2. $pV^2 = \text{constant}$

$$p_{\text{initial}} = 1 \text{ bar} ; V_{\text{initial}} = 0.1 \text{ m}^3 ; p_{\text{final}} = 9 \text{ bar} ; V_{\text{final}} = ???$$

$$(1 \text{ bar}) (0.1 \text{ m}^3)^2 = 0.01 \leftarrow \text{constant}$$

$$\sqrt{V^2} = \frac{\text{constant}}{p}$$

$$V = \sqrt{\frac{\text{constant}}{p}}$$

$$V = \sqrt{\frac{0.01}{9}}$$

$$V = 0.033 \text{ m}^3 \leftarrow \text{Part A}$$

$$W = \int_{V_1}^{V_2} p \, dV \quad \text{we know}$$

$$p = \frac{0.01}{V^2}$$

$$\text{therefore, } W = \int_{V_1}^{V_2} \frac{0.01}{V^2} \, dV \quad \text{part B}$$

$$W = \left(\frac{0.01}{-1} V^{-1} \right) \Big|_{0.1}^{0.33}$$

$$W = -0.01 (0.33^{-1} - 0.1^{-1}) = -203 \text{ bar m}^3 = -20.3 \text{ kJ}$$

1 Question 1 20 / 20

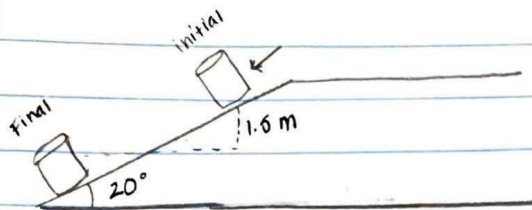
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#a. $pV^2 = \text{constant}$

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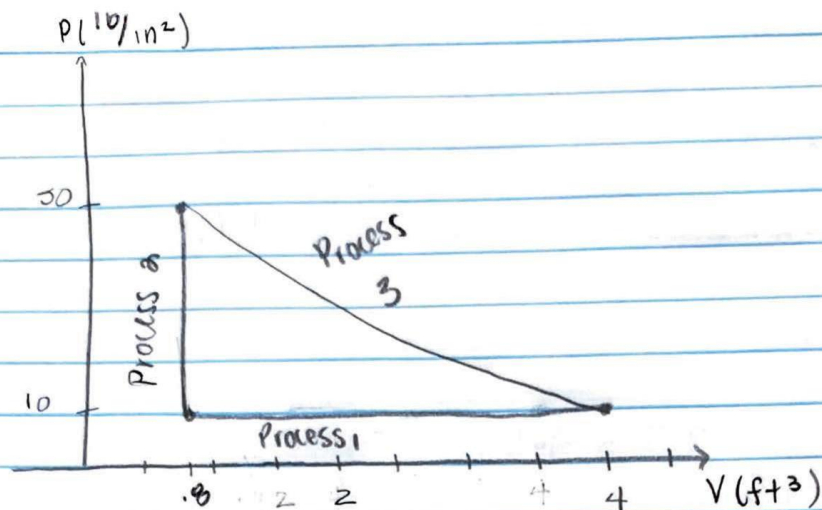
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2 Question 2 20 / 20

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- 0 pts Be more precise on calculation. Keep more significant digits.
- 5 pts Wrong equation in b)
- 5 pts Wrong calculation in a)
- 5 pts Wrong equation in a)

#3.



Part A:

We know $\frac{P_1}{P_2} = \frac{V_2}{V_1} \Rightarrow \frac{10 \text{ lbf/in}^2}{50 \text{ lbf/in}^2} = \frac{V_2}{4 \text{ ft}^3}$

$$V_2 = \frac{10 \cdot 4}{50} = 0.8 \text{ ft}^3$$

Part B:

From Process 1 $\rightarrow$ 2

$$P \cdot V = \text{constant} \Rightarrow 10 \times 4 = 40$$

$$P = \frac{40}{V} \quad \text{then we know } W = \int_{V_1}^{V_2} P dV$$

$$W = \int_{V_1}^{V_2} \frac{40}{V} dV = 40 \ln|V| \Big|_4^{0.8} = 40 (\ln(0.8) - \ln(4)) = -64.4 \text{ ft}^3 \frac{\text{lbf}}{\text{in}^2}$$

$$\text{Conversion } -64.4 \text{ ft}^3 \frac{\text{lbf}}{\text{in}^2} \times 12^3 \times 0.000107 = -11.903 \text{ Btu}$$

Process 2 $\rightarrow$ 3

$$W = 0$$

V is constant therefore no change in volume and No work done

Process 3 $\rightarrow$ 1

$$P_3 = P_1 = 10 \text{ lbf/in}^2$$

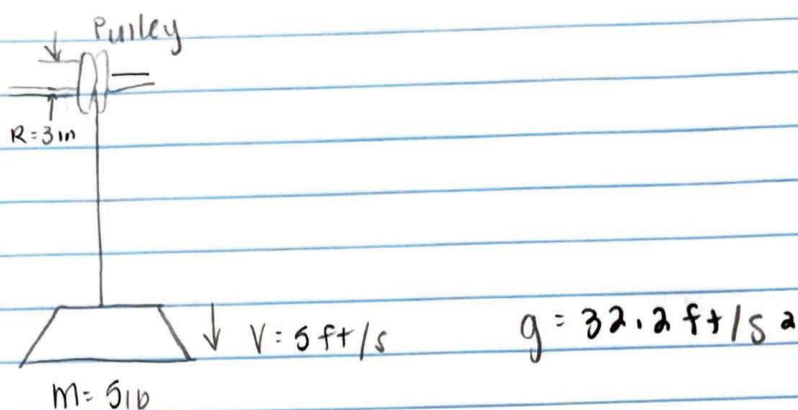
$$W = \int_{V_2}^{V_3} 10 dV = 10V \Big|_{0.8}^4 = 10(4 - 0.8) = 32 \text{ ft}^3 \frac{\text{lbf}}{\text{in}^2} = 5.92 \text{ Btu}$$

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#4.

POWER P

$$P = F \cdot V \quad \& \quad F = mg$$

$$m = \frac{0}{32.2} \text{ lb} \cdot \text{s}^2/\text{ft} ; \quad g = 32.2 \text{ ft/s}^2 ; \quad V = 5 \text{ ft/s}$$

$$W = \left(\frac{5}{32.2} \right) (32.2) (5) = 25 \text{ lbf} \cdot \text{ft/s}$$

$$25 \text{ lbf} \cdot \text{ft/s} \times \left(\frac{1 \text{ hp}}{550 \text{ lbf} \cdot \text{ft/s}} \right) = 0.0455 \text{ hp}$$

Torque τ

$$\tau = .5 \cdot m \cdot g \cdot D$$

$$\tau = .5 \times 5 \times \left(\frac{32.2}{32.2} \right) \times .25 = 0.625 \text{ lbf} \cdot \text{ft}$$

Rotational speed

$$N = \frac{60 P}{2\pi \cdot \tau} = \frac{60 \times 25 \text{ lbf} \cdot \text{ft/s}}{2\pi \times 0.625 \text{ lbf} \cdot \text{ft}} = 301.97 \text{ RPM}$$

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CHE 201

Homework 2

Question 5

Batteries are used on an everyday basis by all individuals. They are the source of power by many devices including but not limited to automobiles, calculators, clocks, etc.. Batteries are composed of many heavy metals which include lead, mercury, cadmium and nickel. These harsh chemicals can be very dangerous to the environment and human beings when disposed incorrectly. After being disposed incorrectly batteries usually result in landfills which then are most likely to discharge chemicals into the soil or water bodies and into the air which pose a threat to the environment. Batteries fall under the category of universal waste as well as some other materials which must be disposed properly. Federal regulations for the disposal of batteries exist which include the promotion of collecting and recycling universal waste. These regulations require the universal waste to be kept in a way where there is no harm to the environment being done. They must then be labeled and taken to a recycling facility in which they properly discard of these items. In the state of Illinois, alkaline batteries are not considered hazardous but are encouraged to be recycled since they can be harsh on the environment. They are often public places that encourage the recycling of batteries by having certain bins in which you may dispose of them in. There are also mail back programs available that make recycling batteries more convenient and it helps the environment. Many auto shops also take in old car batteries if the customer wants a discount on the new battery being bought and that encourages the recycling of this universal waste.

References

“Battery Recycling.” *Ecology Action Center*, ecologyactioncenter.org/hhw/battery-recycling/.

“Universal Waste.” *EPA*, Environmental Protection Agency, 21 Oct. 2019,
www.epa.gov/hw/universal-waste.

Wieman, Bethany. “Importance of Recycling Batteries.” *Home Guides | SF Gate*, 7 Oct.
2016, homeguides.sfgate.com/importance-recycling-batteries-79299.html.

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